Ghosal Hematodiaphyseal Dysplasia (GHDD): A Comprehensive Disease Dossier

Disease: Ghosal Hematodiaphyseal Dysplasia (GHDD / Ghosal syndrome / diaphyseal dysplasia–anemia syndrome) **Causal gene: TBXAS1** (thromboxane A synthase 1; cytochrome P450 CYP5A1) **Identifiers:** MONDO:0009274 · OMIM:231095 · Chromosome 7q33–q34 · Category: Mendelian (autosomal recessive)

Summary

Ghosal Hematodiaphyseal Dysplasia (GHDD) is a rare autosomal-recessive disorder caused by **biallelic loss-of-function mutations in TBXAS1**, the gene encoding **thromboxane A synthase (TXAS, also called CYP5A1)**. Although the disease name points only to the "hematodiaphyseal" phenotype, the causal gene is unambiguously TBXAS1 — identified in consanguineous GHDD families by Geneviève et al. in 2008 (PMID: 18264100) — and this report follows the MONDO ID (MONDO:0009274) and OMIM entry (231095) given in the research template. TXAS converts the cyclooxygenase (COX) product prostaglandin H2 (PGH2) into thromboxane A2 (TXA2); when TXAS is lost, PGH2 is shunted toward other prostaglandins (PGE2, PGD2, PGF2α) and the downstream eicosanoid balance that governs bone remodeling and hematopoiesis is disrupted.

Clinically, GHDD presents in **early childhood** with a characteristic triad: (1) **moderate-to-severe, hematinic-refractory, myelophthisic anemia** (often transfusion-dependent), (2) **increased cortical density and loss of normal constriction of the long-bone diaphyses** (diaphyseal dysplasia/osteosclerosis), and (3) **bone-marrow fibrosis** with increased reticulin and dyserythropoiesis. Splenomegaly, bone pain, and short stature are common. A defining and clinically crucial feature is the **dramatic responsiveness of both the anemia and the skeletal disease to corticosteroids** — a phenomenon first described by Ghosal and colleagues in 1988 (PMID: 3385529), two decades before the gene was known. More recently, **COX-inhibiting NSAIDs (aspirin, ibuprofen)** have emerged as a rational, mechanism-based targeted therapy (PMID: 36574346).

This dossier synthesizes eight confirmed findings and 31 reviewed papers into a full 15-section disease characterization. The strongest and best-supported conclusions are the gene–disease relationship, the prostaglandin-shunting mechanism linked to RANKL/OPG dysregulation, the steroid/NSAID-responsive clinical course, and the faithful biochemical recapitulation of the defect in TXAS-knockout mice. The newest frontier is a phenotypic expansion toward a **COX-pathway-responsive autoinflammatory / chronic nonbacterial osteomyelitis (CNO) spectrum**, suggesting GHDD is better conceived as part of a continuum of eicosanoid-driven inflammatory bone disease.

Key Findings

Finding 1 — GHDD is caused by biallelic loss-of-function mutations in TBXAS1

The genetic basis of GHDD was established by Geneviève et al. (*Thromboxane synthase mutations in an increased bone density disorder (Ghosal syndrome)*, Nat Genet 2008), who identified homozygous **TBXAS1** mutations in consanguineous GHDD families: *"we identified mutations in TBXAS1, which encodes thromboxane synthase (TXAS)"* (PMID: 18264100). TBXAS1 maps to chromosome **7q33–q34** (originally localized to 7q34–q35 by FISH, PMID: 8325653); its HGNC symbol is *TBXAS1* (thromboxane A synthase 1), and the encoded enzyme is a cytochrome P450 (CYP5A1). Inheritance is **autosomal recessive**: affected individuals are homozygous (in consanguineous families) or compound heterozygous (in non-consanguineous families). A 2024 review confirms: *"It is due to biallelic variants in the TBXAS1 (OMIM*274180) gene, which encodes for thromboxane synthase" *(PMID: 39220787). Recurrent pathogenic alleles include p.Arg412Gln (c.1235G>A)** and glycine substitutions at codons 473/474 (p.Gly473/474Trp).

Finding 2 — Mechanism: TXAS loss shunts PGH2 to other prostaglandins and dysregulates osteoblast RANKL/OPG

Thromboxane synthase converts the COX-derived endoperoxide **prostaglandin H2 (PGH2)** into **thromboxane A2 (TXA2)**. Loss of function therefore increases the pool of PGH2 available to competing prostaglandin synthases, raising production of PGE2, PGD2, and PGF2α: *"Loss-of-function mutations in TBXAS result in an increase in PGH2 availability for other PG synthases"* (PMID: 36574346). Critically, Geneviève et al. demonstrated that this eicosanoid pathway feeds directly into bone remodeling: *"We also found that TXAS and TXA(2) modulated expression of*

TNFSF11 and TNFRSF11B (encoding RANKL and osteoprotegerin (OPG), respectively) in primary cultured osteoblasts" (PMID: 18264100). Dysregulation of the **RANKL/OPG axis** — the master switch controlling osteoclastogenesis — provides the mechanistic link between a platelet/eicosanoid enzyme defect and the paradoxical **increase in bone density** seen in GHDD. Patient platelets show a specific deficit in arachidonic-acid-induced aggregation, confirming the enzymatic block in vivo.

Finding 3 — Clinical phenotype: steroid-responsive refractory anemia + diaphyseal osteosclerosis + marrow fibrosis

GHDD classically presents in **early childhood** with moderate-to-severe normocytic, myelophthisic anemia (frequently transfusion-dependent), splenomegaly, bone pain, and radiographic cortical thickening with increased density of the long-bone diaphyses and loss of normal diaphyseal constriction. The marrow shows **fibrosis (increased reticulin)** and dyserythropoiesis. Sharma et al. describe patients who *"presented in early childhood with moderate to severe anemia, splenomegaly, and a hypocellular marrow with increased reticulin"* and note that *"All cases dramatically responded to oral steroid and no longer needed blood transfusion"* (PMID: 25374284). While childhood onset is typical, adult and middle-aged presentations are well documented (PMID: 31395426; PMID: 39220787).

Finding 4 — Treatment: corticosteroids are standard; COX-inhibiting NSAIDs are mechanism-based targeted therapy

Low-dose **corticosteroids** are the standard of care, normalizing hemoglobin, reducing or eliminating transfusion requirements, and halting progression of bony disease (PMID: 25374284; PMID: 27156553). Building on the prostaglandin-shunting mechanism, Brown et al. (2023) treated an adult and a child with standard-dose NSAIDs (aspirin/ibuprofen), reasoning that *"nonsteroidal anti-inflammatory drugs (NSAIDs), which inhibit COX-1 and COX-2, could ameliorate the effects of TBXAS1 loss"*; *"Both patients had rapid improvements concerning hematologic parameters and inflammatory markers without adverse events"*, with mass spectrometry confirming reduced COX and LOX products (PMID: 36574346). In an adult, aspirin 150 mg/day raised hemoglobin from 6.1 to 11 g/dL (PMID: 39220787).

Finding 5 — Model organism: TXAS-knockout mouse recapitulates the biochemical defect

Yu et al. (2004) generated **TXAS(−/−) mice**: *"Platelets from TXAS(-/-) mice failed to aggregate or generate thromboxane B(2) in response to arachidonic acid (AA) but produced increased prostaglandin-E(2) (PGE(2)), PGD(2), and PGF(2 alpha)"* (PMID: 15010374). The knockout had normal platelet counts and megakaryocyte numbers but a **mild hemostatic defect** and **resistance to arachidonate-induced shock**. This cleanly reproduces the human biochemical shunting defect (loss of TXA2, excess of competing prostaglandins) seen in GHDD patient platelets (PMID: 18264100). The model's chief limitation is that it was not reported to reproduce the human osteosclerosis/marrow-fibrosis phenotype.

Finding 6 — Variable expressivity and the TBXAS1 variant spectrum

Reported pathogenic TBXAS1 variants are biallelic and predominantly loss-of-function, spanning **missense** (e.g., c.1235G>A p.Arg412Gln; c.1417G>T/c.1420G>T p.Gly473/474Trp; c.1238G>A p.Arg413Glu; c.266T>C; c.989T>C) and **frameshift** (c.583_584del p.Ala195Leufs*12) alleles (PMID: 18264100; PMID: 33595912; PMID: 28868793; PMID: 27156553). Kim et al. reported three compound-heterozygous siblings with identical variants (*"two heterozygous variants in TBXAS1, exon7, c.583_584del, p.Ala195Leufs*12, and exon12, c.1420G>T, p.Gly474Trp"*) in whom two brothers had transfusion-dependent anemia while their sister had normal blood counts and only radiographic cortical thickening — concluding these *"cases illustrate the variable clinical expressivity of the GHDD"* (PMID: 33595912).

Finding 7 — Phenotypic expansion: a COX-responsive autoinflammatory / CNO spectrum

The most recent literature reframes TBXAS1 deficiency as part of an **eicosanoid-driven autoinflammatory spectrum**. Kisla Ekinci et al. (2025) identified TBXAS1 variants in a child with multifocal **chronic nonbacterial osteomyelitis (CNO)** responsive to NSAIDs, plus two siblings with CNO lacking osteosclerosis or hematologic involvement; *"These findings suggest a potential link between TBXAS1 deficiency and a distinct, milder CNO phenotype potentially mediated by the COX pathway"* (PMID: 41148195). A 2026 case report frames the disorder systemically — *"GHDS is a systemic inflammatory syndrome that responds to anti-inflammatory therapy"* — with elevated CRP (38.9 mg/L) and ESR (80 mm/h) (PMID: 42059562). A 2025 osteosclerotic-disorders cohort found GHDD in 3 of 34 patients across 23 families (PMID: 40198394).

Finding 8 — The original 1988 description established steroid responsiveness and skeletal reversibility

Ghosal et al. (1988) first delineated the syndrome in five early-childhood patients with moderate-to-marked anemia and thick long bones showing radiologic diaphyseal dysplasia. *"the anemia was persistent and not responsive to hematinics, prednisolone was administered to two of these patients and caused remarkable improvement of the anemia in both"*, and *"Roentgenologic examination carried out after 1 year of corticosteroid therapy in one patient and after 11 years in another showed considerable improvement of the bony changes"* (PMID: 3385529). This established both steroid responsiveness and **reversibility of the skeletal phenotype** two decades before TBXAS1 was identified.

The 15-Section Disease Characterization

1. Disease Information

GHDD is a rare Mendelian (autosomal-recessive) skeletal dysplasia–bone marrow failure syndrome defined by the triad of **diaphyseal dysplasia of the long bones, bone-marrow fibrosis, and steroid-responsive refractory anemia**. Key identifiers: **MONDO:0009274**, **OMIM:231095**, gene **TBXAS1** (OMIM *274180). Orphanet lists it among rare primary bone dysplasias with increased bone density. There is no specific ICD-10 code (typically coded under other/unspecified osteosclerosis or diaphyseal dysplasia, e.g., Q78.x). Synonyms: Ghosal syndrome; Ghosal hematodiaphyseal dysplasia; diaphyseal dysplasia–anemia syndrome; hematodiaphyseal dysplasia Ghosal type. The knowledge base for this disease is almost entirely aggregated case-level literature** (individual case reports and small case series) rather than large EHR cohorts, reflecting its rarity.

2. Etiology

Primary cause — genetic: Biallelic loss-of-function variants in **TBXAS1** (PMID: 18264100). **Genetic risk factors:** carriage of two pathogenic TBXAS1 alleles; **consanguinity** is a major risk factor (homozygous cases cluster in consanguineous families, especially from South Asia and the Middle East). **Environmental risk factors:** none identified — the disease is monogenic, not driven by toxins, lifestyle, age, or sex. **Protective factors:** no genetic or environmental protective alleles are described; the only "protective" intervention is pharmacologic

(corticosteroids, NSAIDs) acting downstream on the dysregulated eicosanoid pathway. **Gene–environment interactions:** none established; however, because the phenotype is mediated by the COX/prostaglandin pathway, pharmacologic COX inhibition functions as a modifiable, mechanism-based lever on disease expression.

3. Phenotypes

Phenotype	Type	HPO suggestion	Onset	Frequency
Refractory/ myelophthisic anemia	Laboratory abnormality / clinical sign	HP:0001903 (Anemia); HP:0004840	Early childhood	Very frequent (core feature)
Diaphyseal dysplasia / increased diaphyseal density	Physical/ imaging sign	HP:0100671 (Diaphyseal dysplasia); HP:0011001 (Increased bone mineral density)	Childhood	Very frequent (core)
Bone-marrow fibrosis / myelofibrosis	Laboratory/ histopathology	HP:0011974 (Myelofibrosis)	Childhood	Frequent
Splenomegaly	Clinical sign	HP:0001744	Childhood	Common
Bone pain	Symptom	HP:0002653	Childhood	Common
Thrombocytopenia / pancytopenia	Laboratory	HP:0001873; HP:0001876 (Pancytopenia)	Variable	Subset (esp. adults)
Short stature	Physical	HP:0004322	Childhood	Variable
Platelet aggregation defect (AA-induced)	Laboratory	HP:0003540 (Impaired platelet aggregation)	Congenital/ latent	Characteristic, often subclinical
Elevated inflammatory markers (CRP/ESR)	Laboratory	HP:0011227 (Elevated CRP)	Variable	Reported in inflammatory presentations

Severity/progression: variable — from transfusion-dependent severe anemia to purely radiographic cortical thickening in minimally symptomatic relatives (PMID: 33595912). Without treatment the anemia is progressive; with steroids/NSAIDs it is largely reversible. **Quality of life:** untreated disease imposes substantial burden (chronic transfusions, iron overload risk, bone pain, impaired growth); timely treatment restores near-normal function and growth (PMID: 39277773).

4. Genetic / Molecular Information

Causal gene: TBXAS1 (HGNC thromboxane A synthase 1; CYP5A1; chromosome 7q33–q34; OMIM *274180; NCBI Gene 6916). Variant classes: missense and frameshift predominate, all loss-of-function; reported alleles include c.1235G>A (p.Arg412Gln), c.1417G>T/c.1420G>T (p.Gly473/474Trp), c.1238G>A (p.Arg413Glu), c.266T>C, c.989T>C, and c.583_584del (p.Ala195Leufs*12). **Classification (ACMG/AMP):** pathogenic/likely-pathogenic for recurrent alleles; some reported as VUS at discovery then reclassified by phenotype/segregation (PMID: 28868793). **Allele frequency:** all are rare in gnomAD, consistent with a recessive disorder. **Origin:** germline. **Functional consequence:** loss of thromboxane synthase activity (enzymatic LOF). **Modifier genes:** none firmly established; variable expressivity among siblings with identical genotypes implies modifiers or stochastic/environmental factors. **Epigenetic / chromosomal abnormalities:** none reported — GHDD is a single-gene disorder without characteristic large structural changes.

5. Environmental Information

No environmental, lifestyle, or infectious contributors cause GHDD; it is a purely monogenic recessive disease. The only environmentally modifiable dimension is **pharmacologic modulation of the eicosanoid pathway** (COX inhibition improves disease). **Infectious agents:** not applicable — the "osteomyelitis" in the expanded CNO phenotype is **non-bacterial/sterile** autoinflammation, not infection (PMID: 41148195).

6. Mechanism / Pathophysiology

Ordered causal chain:

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1. Biallelic LOF mutation in TBXAS1 (7q33-q34)
      → results in loss of thromboxane A synthase (TXAS/CYP5A1) activity.
2. Loss of TXAS activity
      → blocks conversion of PGH2 → thromboxane A2 (TXA2).
3. Block of TXA2 synthesis
      → leads to accumulation/shunting of PGH2 toward other PG synthases.
   ├─(branch A, bone) → excess PGE2/PGD2/PGF2α in osteoblasts
   │       → dysregulates TNFSF11 (RANKL) / TNFRSF11B (OPG) expression
   │       → alters osteoclast/osteoblast balance
   │       → results in increased diaphyseal bone density (osteosclerosis)
   │       → encroaches on marrow space / "haphazard ossification."
   ├─(branch B, marrow) → eicosanoid imbalance + marrow fibrosis (↑reticulin)
   │       → myelophthisic/dyserythropoietic process
   │       → results in refractory anemia ± pancytopenia + splenomegaly.
   └─(branch C, platelets) → loss of TXA2
           → impaired arachidonic-acid-induced platelet aggregation
           → mild bleeding tendency (often subclinical).
4. Downstream pro-inflammatory eicosanoid excess (inferred)
      → systemic inflammation (↑CRP/ESR), sterile osteomyelitis in some
      → defines the autoinflammatory/CNO end of the spectrum.
5. Therapeutic reversal: COX inhibition (steroids act broadly; NSAIDs act
   directly on COX-1/COX-2)
      → reduces PGH2 flux and downstream prostaglandins
      → reverses anemia and halts/reverses bony disease.
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Upstream vs downstream: the TBXAS1 mutation and loss of TXAS are the upstream initiating lesion; prostaglandin shunting is the pivotal node; RANKL/OPG dysregulation (bone), marrow fibrosis (hematopoiesis), and platelet dysfunction are parallel downstream branches. **Molecular pathways:** arachidonic acid / eicosanoid metabolism (KEGG hsa00590), COX (PTGS1/PTGS2) → PGH2 → TXAS → TXA2; RANKL/OPG/RANK signaling in osteoclastogenesis. **Cellular processes:** osteoblast–osteoclast coupling, bone remodeling, hematopoiesis, platelet activation, inflammation. **Protein dysfunction:** enzymatic loss of function of a cytochrome-P450 thromboxane synthase. **Biochemical abnormality:** thromboxane synthase deficiency. Steps 4 (systemic inflammation) and the precise in-vivo route from prostaglandin shunting to osteosclerosis are partly **inferred** rather than fully demonstrated.

GO term suggestions: GO:0019371 (cyclooxygenase pathway), GO:0001516 (prostaglandin biosynthetic process), GO:0046457 (prostanoid biosynthetic process), GO:0030168 (platelet activation), GO:0045453 (bone resorption), GO:0001503 (ossification), GO:0006954 (inflammatory response). **CL term suggestions:** CL:0000062 (osteoblast), CL:0000092

(osteoclast), CL:0000233 (platelet), CL:0000765 (erythroblast), CL:0000057 (fibroblast). **CHEBI:** CHEBI:15035 (prostaglandin H2), CHEBI:15627 (thromboxane A2), CHEBI:15551 (prostaglandin E2), CHEBI:15842 (arachidonic acid).

7. Anatomical Structures Affected

- **Primary organs:** long bones (diaphyses/metadiaphyses) — UBERON:0001474 (bone element), UBERON:0004053 (diaphysis); **bone marrow** — UBERON:0002371.
- **Secondary organ involvement:** spleen (splenomegaly, extramedullary hematopoiesis) — UBERON:0002106.
- **Body systems:** skeletal system, hematopoietic/immune system; subclinical hemostatic (platelet) involvement.
- **Tissue/cell level:** osteoblasts (CL:0000062), osteoclasts (CL:0000092), marrow stromal fibroblasts (CL:0000057), erythroid precursors (CL:0000765), platelets/megakaryocytes (CL:0000233 / CL:0000556).
- **Subcellular:** endoplasmic reticulum–associated cytochrome P450 enzyme (GO:0005789, ER membrane); cytosolic eicosanoid signaling.
- **Localization/lateralization: bilateral, symmetric** long-bone involvement.

8. Temporal Development

- **Onset:** typically **congenital/early childhood** (first years of life), insidious/chronic; adult and middle-aged first presentations occur (PMID: 31395426; PMID: 39220787).
- **Progression:** chronic and progressive if untreated (worsening anemia, advancing sclerosis); **treatment-induced remission** with corticosteroids/NSAIDs. Bony changes are partially reversible over months to years (PMID: 3385529).
- **Course pattern:** chronic lifelong disorder requiring ongoing therapy; not self-limited.
- **Critical period:** early diagnosis and steroid initiation can halt bone damage and avoid chronic transfusion dependence, improving growth and QoL (PMID: 39277773).

9. Inheritance and Population

- **Inheritance:** autosomal recessive (AR); homozygous in consanguineous families, compound heterozygous in non-consanguineous families.
- **Penetrance/expressivity:** apparently high penetrance for the biochemical/skeletal trait but **variable expressivity**, including near-normal hematology in a sibling sharing the genotype (PMID: 33595912).

- **Epidemiology:** ultra-rare; fewer than ~50–100 genetically confirmed cases worldwide. No reliable prevalence/incidence figures exist; in one osteosclerosis cohort GHDD accounted for 3/34 patients (PMID: 40198394).
- **Consanguinity/founder effects:** consanguinity is a strong contributor; recurrent alleles (e.g., p.Arg412Gln, p.Arg413Glu) reported predominantly in South Asian families suggest possible founder/recurrent mutations.
- **Sex ratio:** no sex predilection (AR). **Geographic distribution:** cases reported from South Asia (India), the Middle East, East Asia, and Europe. **Anticipation/mosaicism:** not applicable/not reported.

10. Diagnostics

- **Laboratory:** CBC showing refractory normocytic anemia ± pancytopenia; peripheral smear showing myelophthisic features; elevated inflammatory markers (CRP/ESR) in inflammatory presentations.
- **Bone marrow biopsy:** hypocellular marrow with **increased reticulin/fibrosis** and dyserythropoiesis; haphazard ossification replacing hematopoiesis (PMID: 28868793).
- **Imaging (key):** plain radiographs of long bones showing **cortical thickening, increased diaphyseal density, and loss of normal diaphyseal constriction** — the diagnostic hallmark that should prompt genetic testing in a child with refractory anemia.
- **Specialized:** platelet aggregation studies showing selective **defect in arachidonic-acid-induced aggregation** with preserved response to PGH2/TXA2 analogues (analogous to thromboxane-pathway defects, cf. PMID: 8562397, PMID: 11442478).
- **Genetic testing (confirmatory):** single-gene **TBXAS1 sequencing**, bone-dysplasia/bone-marrow-failure **gene panels**, or **whole-exome sequencing** (PMID: 28868793; PMID: 35395429). CMA/karyotype/FISH are not informative (point mutations).
- **Differential diagnosis:** Camurati–Engelmann disease (TGFB1), craniometaphyseal dysplasia, juvenile Paget disease (JPD-5), other osteosclerotic dysplasias, and other causes of childhood myelofibrosis/aplastic anemia (PMID: 40198394). A rare caveat: TBXAS1 variants may surface during iron-refractory anemia workups (PMID: 37951373).

11. Outcome / Prognosis

With prompt corticosteroid (or NSAID) therapy, prognosis is **generally favorable**: hemoglobin normalizes, transfusion dependence resolves, and bony progression halts or reverses (PMID: 25374284; PMID: 3385529). Untreated disease carries morbidity from chronic anemia,

transfusion-related iron overload, bone pain, and impaired growth. No disease-specific mortality statistics exist given rarity. **Prognostic factors:** timeliness of diagnosis/treatment; severity of marrow fibrosis; genotype–phenotype variability. Life expectancy appears near-normal with treatment, though long-term cohorts are lacking.

12. Treatment

Therapy	Class / target	Effect	Evidence
Corticosteroids (e.g., prednisolone, low dose)	Broad anti-inflammatory; suppresses eicosanoid pathway	Normalizes Hb, ends transfusions, halts/ reverses bone disease	PMID: 25374284, PMID: 3385529, PMID: 27156553
NSAIDs (aspirin, ibuprofen)	COX-1/COX-2 inhibition (mechanism-based)	Rapid hematologic/ inflammatory improvement; reduces COX/LOX products	PMID: 36574346, PMID: 39220787
Blood transfusion	Supportive	Temporizing for severe anemia pre-diagnosis	Multiple case reports

NCIT suggestions: NCIT:C305 (Aspirin), NCIT:C561 (Ibuprofen), NCIT:C683 (Prednisolone), NCIT:C154652 (Corticosteroid therapy), NCIT:C15431 (Blood transfusion). **Pharmacogenomics / gene & cell therapy:** none specific to GHDD. The NSAID strategy is a textbook example of **genotype-guided personalized medicine** — inhibiting COX upstream of the shunted prostaglandin production. **Adverse events:** standard steroid (growth, metabolic) and NSAID (GI, renal) risks; NSAIDs were well tolerated in reported GHDD patients (PMID: 36574346).

13. Prevention

No primary prevention exists for this monogenic disorder. Relevant measures are **genetic**: carrier/cascade testing in affected families, **genetic counseling** for consanguineous couples, and **prenatal/preimplantation diagnosis** where a familial TBXAS1 genotype is known. **Secondary prevention** = early recognition (refractory anemia + diaphyseal cortical thickening → TBXAS1 testing) enabling prompt steroid/NSAID therapy to prevent skeletal damage and transfusion dependence. No immunization or public-health measures apply.

14. Other Species / Natural Disease

- **Orthologous genes:** mouse *Tbxas1* and rat *Tbxas1* encode thromboxane synthase; human *TBXAS1* is NCBI Gene 6916. The eicosanoid/thromboxane pathway is highly conserved across mammals.
- **Natural disease:** no well-characterized naturally occurring GHDD-equivalent reported in companion animals or wildlife (no established OMIA entry for a spontaneous animal GHDD).
- **Comparative biology:** the mouse knockout recapitulates the biochemistry (Finding 5), supporting evolutionary conservation of the mechanism, though the skeletal phenotype is not reproduced. **Zoonotic potential:** not applicable (genetic disease).

15. Model Organisms

- **Primary model: TXAS(−/−) knockout mouse** (Yu et al. 2004, PMID: 15010374) — a genetic (constitutive) knockout.
- **Phenotype recapitulation: Biochemistry: excellent** — loss of TXA2 and compensatory PGE2/PGD2/PGF2α; **hemostasis: partial** (mild bleeding defect, resistance to arachidonate-induced shock); **skeletal/marrow: not reported** — the knockout does not clearly reproduce the human osteosclerosis and marrow fibrosis.
- **Limitations:** species differences in bone remodeling and eicosanoid–RANKL/OPG coupling; absence of a reported skeletal phenotype limits its use for studying the bone disease. **Applications:** ideal for dissecting platelet/eicosanoid biology and testing COX-pathway pharmacology; patient-derived **iPSC/osteoblast** systems (as used functionally by Geneviève et al. for RANKL/OPG) are the complementary model for the bone phenotype.
- **Resources:** MGI (Tbxas1), IMPC/IMSR for mouse lines.

Mechanistic Model / Interpretation

The unifying model is that **GHDD is a single-enzyme eicosanoid-balance disorder**. A recessive loss of thromboxane synthase removes one specific drain on the common precursor PGH2. The resulting back-pressure redirects PGH2 into excess PGE2, PGD2, and PGF2α. Because these prostaglandins are potent local regulators of bone remodeling (via RANKL/OPG in osteoblasts), hematopoiesis, inflammation, and platelet function, a defect that looks "just like" a platelet thromboxane deficiency instead produces a dramatic multisystem phenotype

dominated by **osteosclerosis** and **marrow failure**. The therapeutic logic closes the loop elegantly: inhibit COX upstream (NSAIDs, or broadly, steroids) → reduce total PGH2 flux → lower the excess downstream prostaglandins → reverse the bone and marrow disease. The newest data (CNO/autoinflammatory spectrum, elevated CRP/ESR) suggest the excess prostaglandins/leukotrienes also drive **sterile inflammation**, placing GHDD on a continuum with eicosanoid-driven autoinflammatory bone disease.

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TBXAS1 LOF ─► ↓TXAS ─► PGH2 shunt ─┬─► ↑PGE2/D2/F2α ─► RANKL/OPG shift ─► osteosclerosis
                                   ├─► marrow fibrosis ─► refractory anemia
                                   ├─► ↓TXA2 ─► platelet aggregation defect
                                   └─► pro-inflammatory eicosanoids ─► CNO / ↑CRP/ESR
          COX inhibition (NSAID/steroid) ─◄──────── reverses the whole cascade
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Evidence Base

PMID	Title (abbrev.)	Role in this report
18264100	Thromboxane synthase mutations in Ghosal syndrome	**Landmark** — gene discovery; RANKL/OPG mechanism
36574346	NSAIDs as targeted therapy for GHDD	Mechanism-based NSAID treatment; PGH2 shunting
15010374	TXAS-deleted mice	Model organism; biochemical recapitulation
3385529	Diaphyseal dysplasia with anemia (Ghosal 1988)	Original description; steroid response, reversibility
25374284	Steroid-responsive anemia in GHDD	Clinical triad; steroid responsiveness
33595912	Novel compound-het TBXAS1 variants	Variant spectrum; variable expressivity
41148195	TBXAS1 deficiency as monogenic CNO	Phenotypic expansion; COX-pathway CNO
42059562	GHDD case report & review	Systemic inflammatory framing; ↑CRP/ESR
40198394	Large osteosclerotic-disorders cohort	Epidemiology; differential diagnosis
39220787	Middle-aged Ghosal syndrome	Adult onset; aspirin efficacy; biallelic confirmation
28868793	Steroid-response pancytopenia, TXAS deficiency	Adult pancytopenia; genetic-diagnosis-driven therapy
27156553	Ghosal type HDD (p.Arg413Glu)	Recurrent variant; low-dose steroid
31395426	Adult siblings with autoimmune anemia	Adult/atypical presentation
8325653	TBXAS1 mapping to 7q34-q35	Gene localization
37951373	IRIDA genomics series	TBXAS1 in refractory-anemia differential
35395429	Novel TBXAS1 variants, Indian children	Variant spectrum; steroid responsiveness
39277773	Steroid-responsive anemia with bony dysplasias	Clinical presentation; QoL benefit of early Rx

Supporting context on thromboxane-pathway platelet defects: PMID: 8562397, PMID: 11442478, PMID: 20357745.

Limitations and Knowledge Gaps

1. **Sparse epidemiology:** no reliable prevalence/incidence; evidence is almost entirely individual case reports and small series — subject to publication and ascertainment bias.
2. **Incomplete bone mechanism:** the RANKL/OPG link is demonstrated in cultured osteoblasts, but the full causal chain from prostaglandin shunting to diaphyseal osteosclerosis in vivo remains partly inferred. The mouse knockout does not reproduce the skeletal phenotype, leaving a model gap.
3. **Genotype–phenotype relationships** are poorly resolved; modifiers explaining variable expressivity (e.g., unaffected siblings) are unidentified.
4. **No controlled treatment trials:** steroid and NSAID efficacy rest on case-level observation; optimal drug, dose, duration, and long-term safety are undefined.
5. **Boundaries of the phenotypic spectrum** (classic GHDD vs. isolated CNO vs. systemic autoinflammation) are still being mapped.
6. **No long-term natural-history cohorts**, so adult outcomes and late complications are uncertain.

Proposed Follow-up Experiments / Actions

1. **Targeted eicosanoid profiling** (LC-MS/MS of urinary/serum prostaglandin and thromboxane metabolites) in TBXAS1 patients before/after NSAID therapy to validate the shunting biomarker and establish a treatment-response readout.
2. **Bone-specific model:** generate a conditional/osteoblast-targeted Tbxas1 knockout, or differentiate patient iPSCs into osteoblasts/osteoclasts, to reproduce and dissect the osteosclerosis and RANKL/OPG mechanism in vivo.
3. **Genotype–phenotype & modifier study:** aggregate a multi-center TBXAS1 registry to correlate variant type (missense vs. truncating) with hematologic vs. skeletal vs. inflammatory severity and search for modifier loci.

4. **Prospective treatment comparison:** structured observational protocol comparing low-dose corticosteroid vs. NSAID (aspirin/ibuprofen) on hemoglobin, transfusion need, inflammatory markers, and radiographic bone changes, with predefined safety monitoring.
5. **Spectrum delineation:** systematically screen TBXAS1 in pediatric CNO/CRMO and unexplained childhood myelofibrosis cohorts to define the true breadth and frequency of TBXAS1-related disease.
6. **Mechanistic immunology:** profile the pro-inflammatory eicosanoid/leukotriene milieu to test whether the autoinflammatory (CNO, ↑CRP/ESR) phenotype is driven by specific downstream prostaglandins amenable to selective inhibition.

Report compiled from 8 confirmed findings and 31 reviewed papers across a 5-iteration autonomous investigation. Evidence is predominantly human clinical (case reports/series) with one mouse-model and in-vitro osteoblast mechanistic study.
